

PRACTICAL_5

Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
srivalli@DESKTOP-6S9T6SD: ~$ mkdir OSLab
mkdir: cannot create directory 'OSLab': File exists
srivalli@DESKTOP-6S9T6SD: ~$ cd OSLab
srivalli@DESKTOP-6S9T6SD: ~/OSLab$ mkdir Practical5
srivalli@DESKTOP-6S9T6SD: ~/OSLab$ cd Practical5
srivalli@DESKTOP-6S9T6SD: ~/OSLab/Practical5$ nano producer_consumer.c
srivalli@DESKTOP-6S9T6SD: ~/OSLab/Practical5$ gcc producer_consumer.c -o producer_consumer
srivalli@DESKTOP-6S9T6SD: ~/OSLab/Practical5$ ./producer_consumer
Producer: Data produced by parent process
Communication time: 0.000009 seconds
Consumer received: Data produced by parent process
srivalli@DESKTOP-6S9T6SD: ~/OSLab/Practical5$ |
```

```
srivalli@DESKTOP-6S9T6SD: ~/OSLab/Practical5$ nano ls_grep.c
srivalli@DESKTOP-6S9T6SD: ~/OSLab/Practical5$ gcc ls_grep.c -o ls_grep
srivalli@DESKTOP-6S9T6SD: ~/OSLab/Practical5$ ./ls_grep
-rw-r--r-- 1 srivalli srivalli 1008 Aug 29 14:59 ls_grep.c
-rwxr-xr-x 1 srivalli srivalli 16400 Aug 29 14:57 producer_consumer
-rw-r--r-- 1 srivalli srivalli 1030 Aug 29 14:57 producer_consumer.c
srivalli@DESKTOP-6S9T6SD: ~/OSLab/Practical5$ ls -l | grep ".c"
-rw-r--r-- 1 srivalli srivalli 1008 Aug 29 14:59 ls_grep.c
-rwxr-xr-x 1 srivalli srivalli 16400 Aug 29 14:57 producer_consumer
-rw-r--r-- 1 srivalli srivalli 1030 Aug 29 14:57 producer_consumer.c
srivalli@DESKTOP-6S9T6SD: ~/OSLab/Practical5$ ls
ls_grep  ls_grep.c  producer_consumer  producer_consumer.c
```

